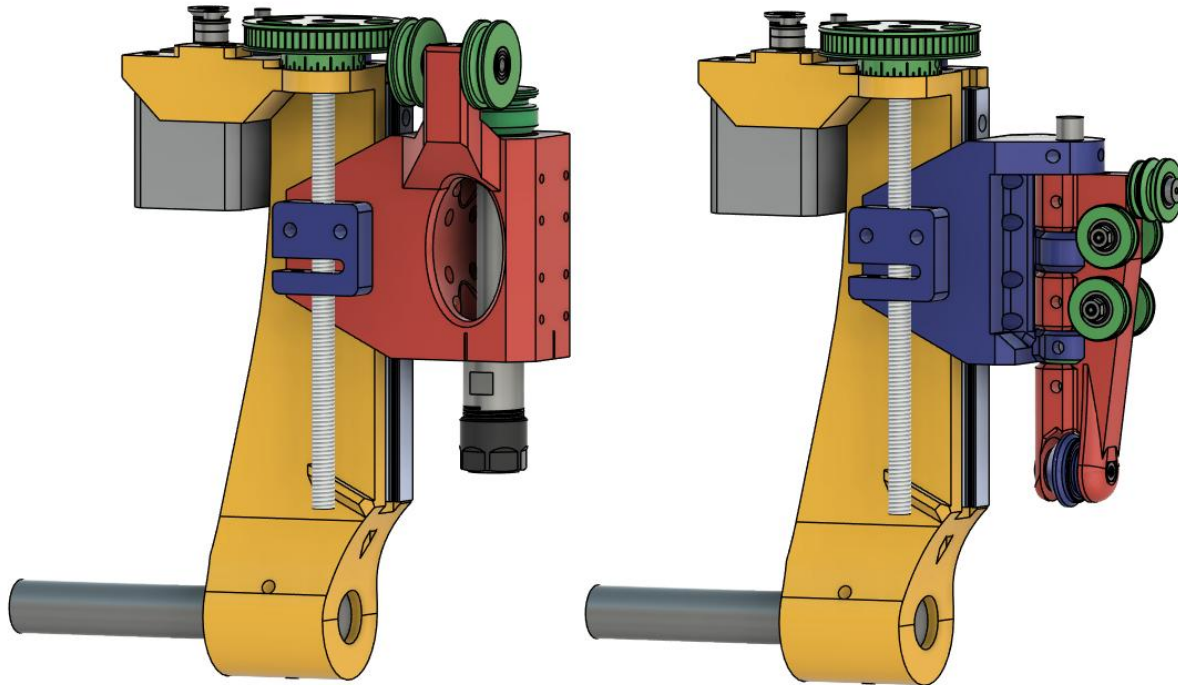


Multi Angle Arm

With

Drilling Frame and Universal Cutting Frame



The MultiAngleArm (MAA) is designed to hold a cutter above the workpiece at any angle between 90 and 180 degrees. The drilling frame rotates the tool parallel to the arm while the universal cutting frame rotates the tool perpendicular to the arm and can be positioned at an angle to the arm. The stepper motor allows the depth of cut to be set programmatically as well as moving the cutter in and out of the workpiece at the beginning and end of a cut. The MAA is attached to the toolpost with an AXA boring bar holder.

Parts List

Part Name	Quantity	Material
Main Arm		
Multi_Angle_Arm	1	ABS-GF
Modifier_TF	1	ABS-GF
Shaft 0.75"x4"	1	
MGN12x100mm Rail	1	

MGN12C Carriage	2	
IdlerAdjuster	1	ABS-GF
F604ZZ Bearing	2	
M604ZZ Bearing	2	
Leadscrew T8x150mm	1	
TR8x2 AntiBacklash Nut	1	
Leadscrew Lock	1	ABS,Brass, or Steel
HTD-3-60t	1	ABS-GF
Stepper_NEMA17	1	
GT2-3-15t or HTD-3-15t	1	Aluminum or ABS-GF
GT2-3 or HTD-3 Timing Belt 80T, 240L,6mm	1	
M3 Square Nut	3	
M4 Square Nut	2	
M5 Square Nut	5	
M3x6 Set Screw	4	
M5x6 Set Screw	3	
M3x8 SHCS	4	
M3x10 SHCS	6	
M4x10 SHCS	2	
M5x12 SHCS	2	
M4x10 Shoulder Bolt (M3x6mm thread)	2	

Drilling Frame

DrillFrame_Housing	1	ABS-GF
IdlerPulley M695ZZ	2	ABS-GF
Spacer_Idler 5x7.5x1.5	1	ABS,Brass, or Steel
M695 bearing	2	
M4 Square Nut	2	
M3x6 Set Screw	2	
M3x8 SHCS	8	
M5x10 Shoulder Bolt	2	
C12-ER16-L100	1	
MR6201ZZ Bearing	2	
Pulley_12mm_Bore	1	ABS-GF

C20-ER25-L100 or C20-ER16-100L	1	
MF6804ZZ Bearing	2	
Pulley_20mm_Bore	1	ABS-GF

UCF

UCF_Base_05c	1	ABS-GF
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UCF_Top	1	ABS-GF
Idler_Pulley_M695ZZ	4	ABS-GF
M695ZZ Bearing	6	
DoublePulley	1	ABS-GF
Spacer_DoublePulley 5x7x4.3	1	ABS,Brass, or Steel
Shaft_10x100	1	
BearingShim_5x7	2	
Round Poly Belt 2x300	1	Polypropylene
M4_Nut	2	
M4 Square Nut	2	
M4_Washer	2	
M5x6 Set Screw	8	
M5x25 Shoulder Bolt	3	
M5x20 Shoulder Bolt	1	

Cutter Heads

CutterHead_TDHB071202_CW_Nut	1	ABS
Spacer_5x7x6.1	1	ABS,Brass, or Steel
M695ZZ Bearing	2	
TDHB07T1202	1	Carbide
Insert_Top_Lock	1	ABS
M2.2x0.45 Torx Head Screw	1	
#2-56 Nut	1	Brass

CutterHead_TDHB071202_CCW_Nut	1	ABS
Spacer_5x7x6.1	1	ABS,Brass, or Steel
M695ZZ Bearing	2	
TDHB07T1202	1	Carbide
Insert_Top_Lock	1	ABS
M2.2x0.45 Torx Head Screw	1	
#2-56 Nut	1	Brass

CutterHead_Round	1	ABS
Spacer_5x7x6.1	1	ABS,Brass, or Steel
M695ZZ Bearing	2	
0.125" diameter cutter	1	Carbide
M3x6 Set Screw	2	

Sharpening Jig 60 degree	1	ABS
Sharpening Jig 90 degree	1	ABS
Sharpening Jig 120 degree	1	ABS

Tools

M2.2x0.45 Tap (UCF only)

M1.75 Drill (UCF only)

M3x0.5 Tap

M2.5 Drill

0.125" Drill (UCF only)

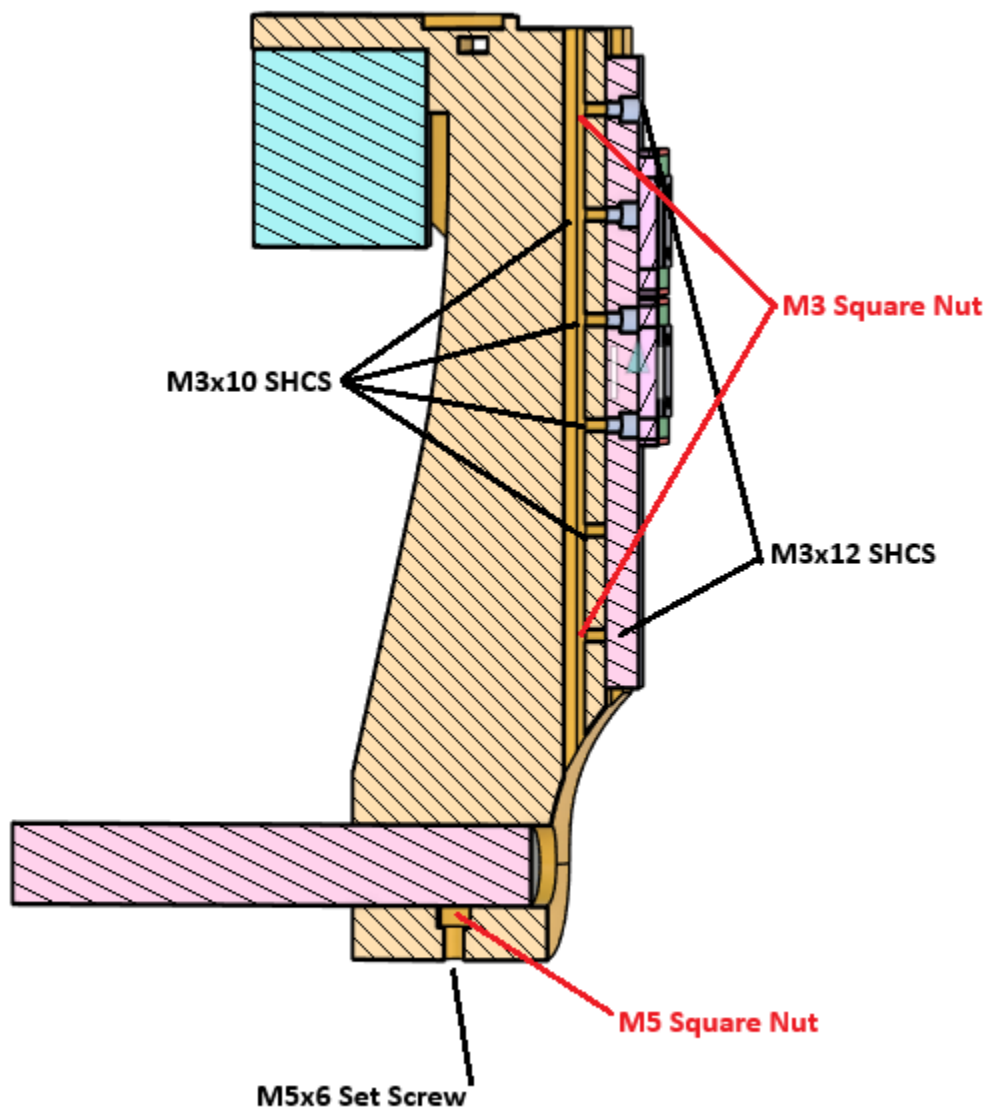
Metric Allen Wrenches

Notes:

When there are multiple options, the preferred option is highlighted in light green and the others in light orange.

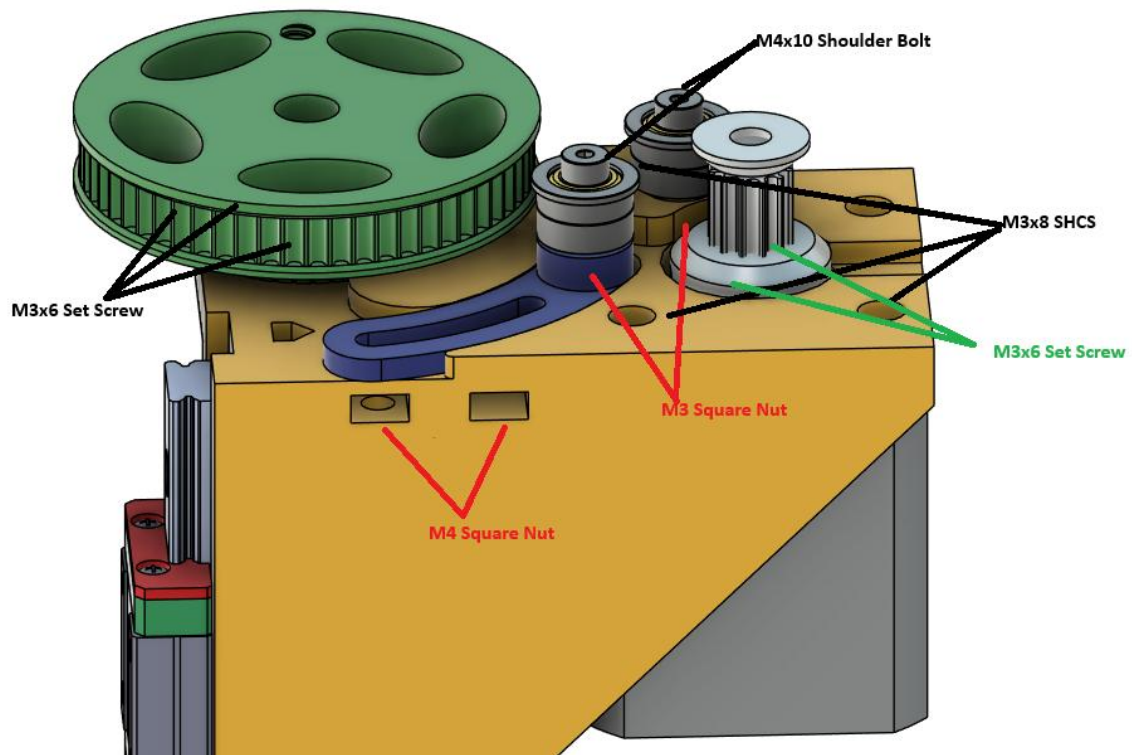
Shoulder bolt threads should be 5 to 6mm long.

Main Arm Assembly



Tap the center 4 holes for M3x0.5. Insert M3 square nuts for the two end screws. When installing the MGN12 rail, leave the rubber bumpers in the end holes until the drilling frame or UCF is installed to prevent the carriages from coming off the rail.

Insert the three M5 square nuts and M5 set screws to lock the shaft.

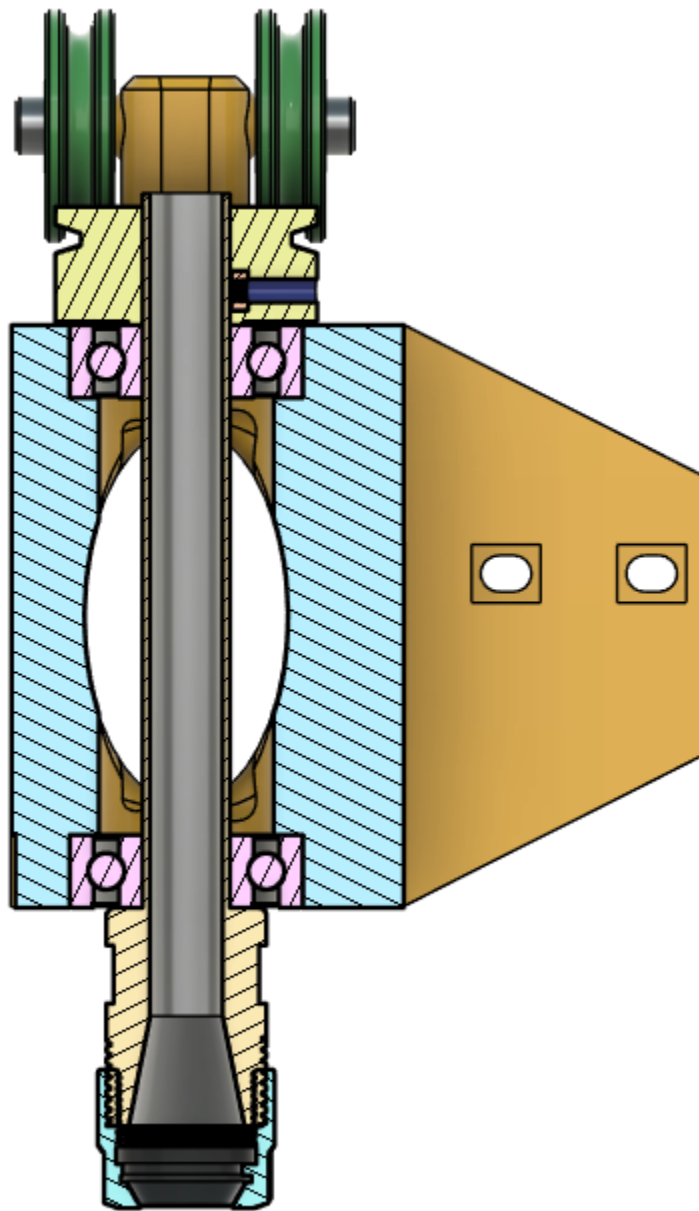


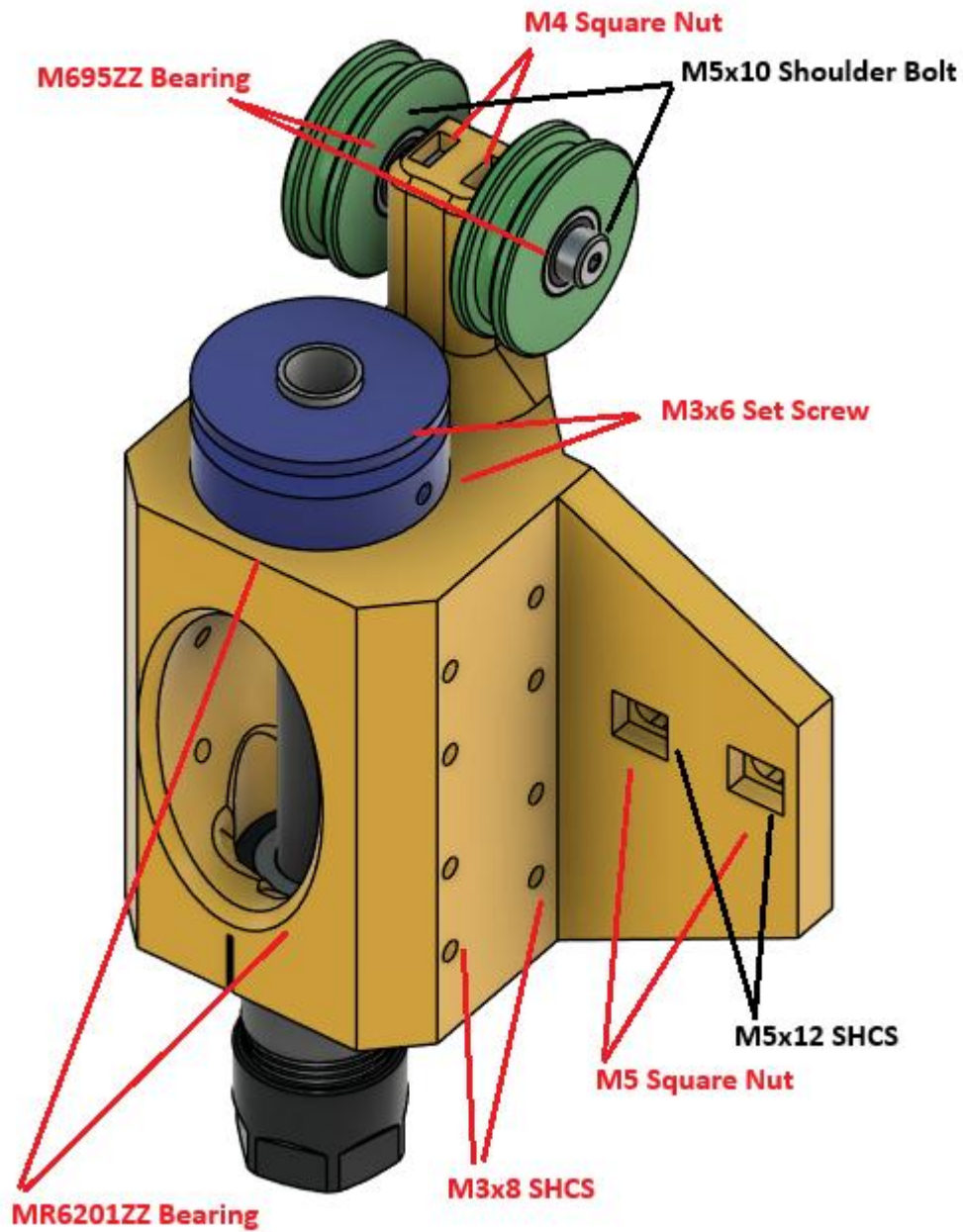
Attach M604_ZZ and F604_ZZ bearings using M4x10 shoulder bolts and M3 square nuts.
Attach adjustable idler using two M4x10 SHCS and M4 washers.

Attach stepper motor using M3x8 SHCS.

Attach Pulley using three M3x6 set screws.

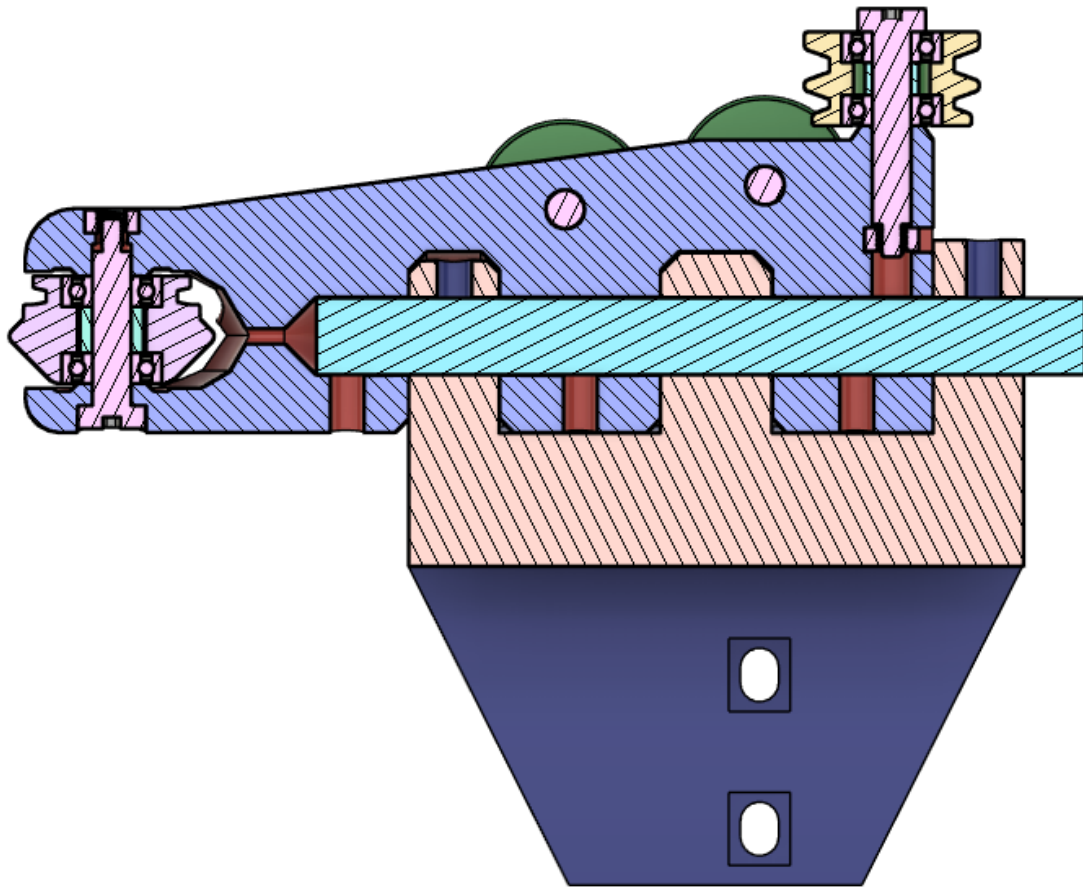
Drilling Frame

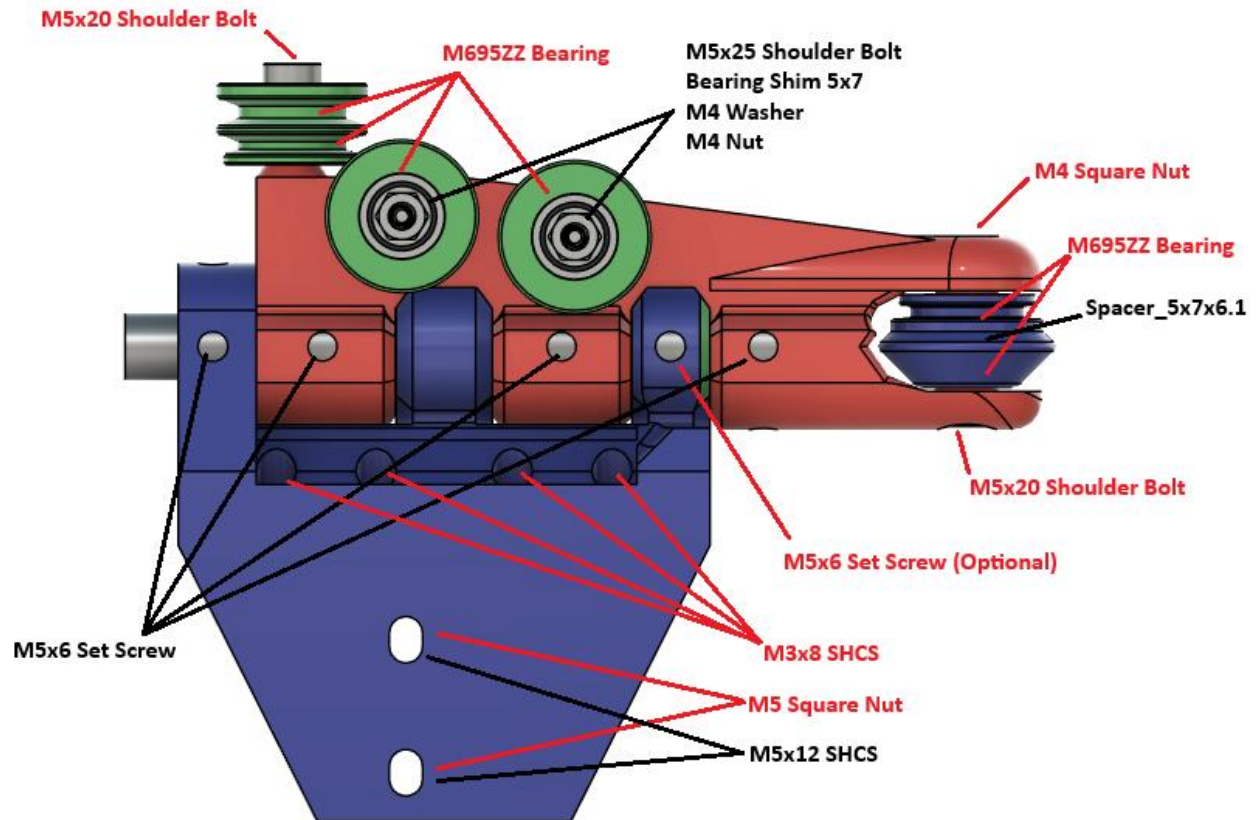




Tap the double pulley with M3 tap.

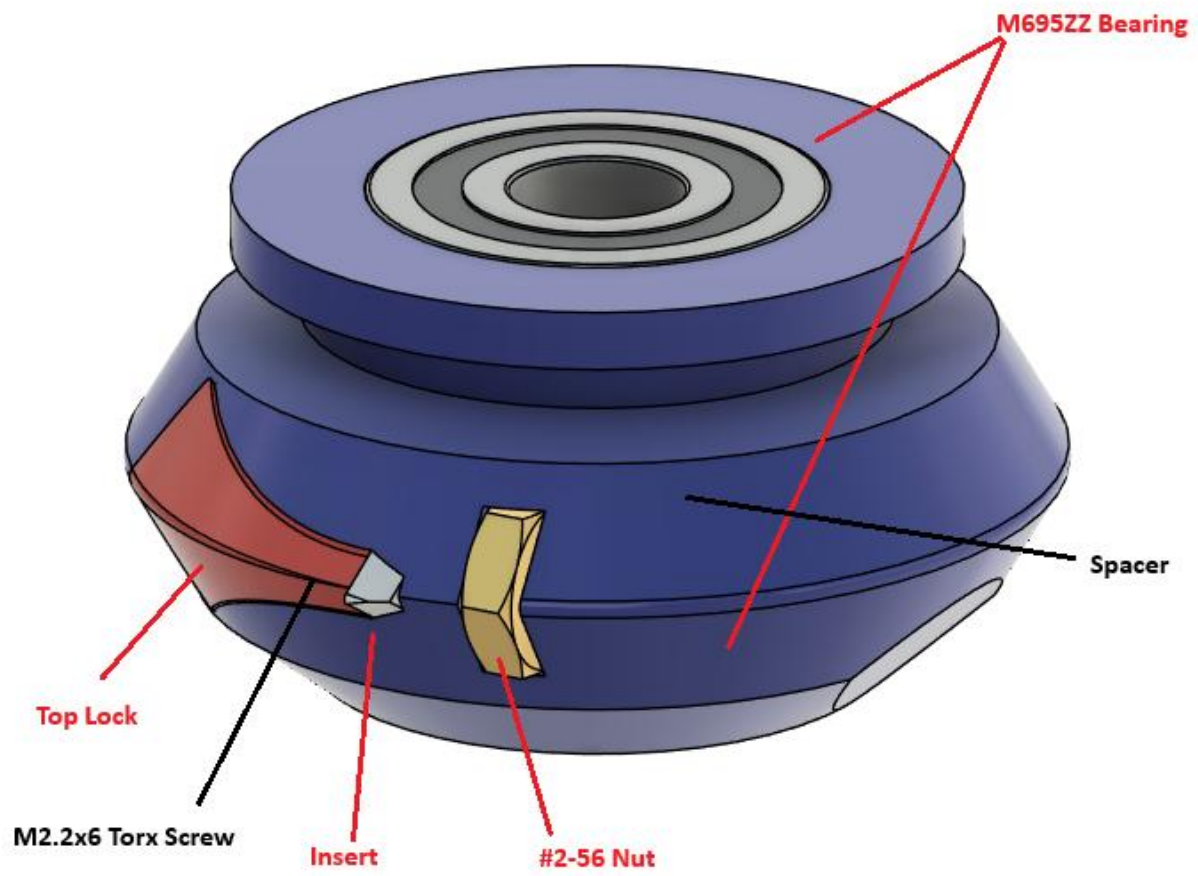
UCF





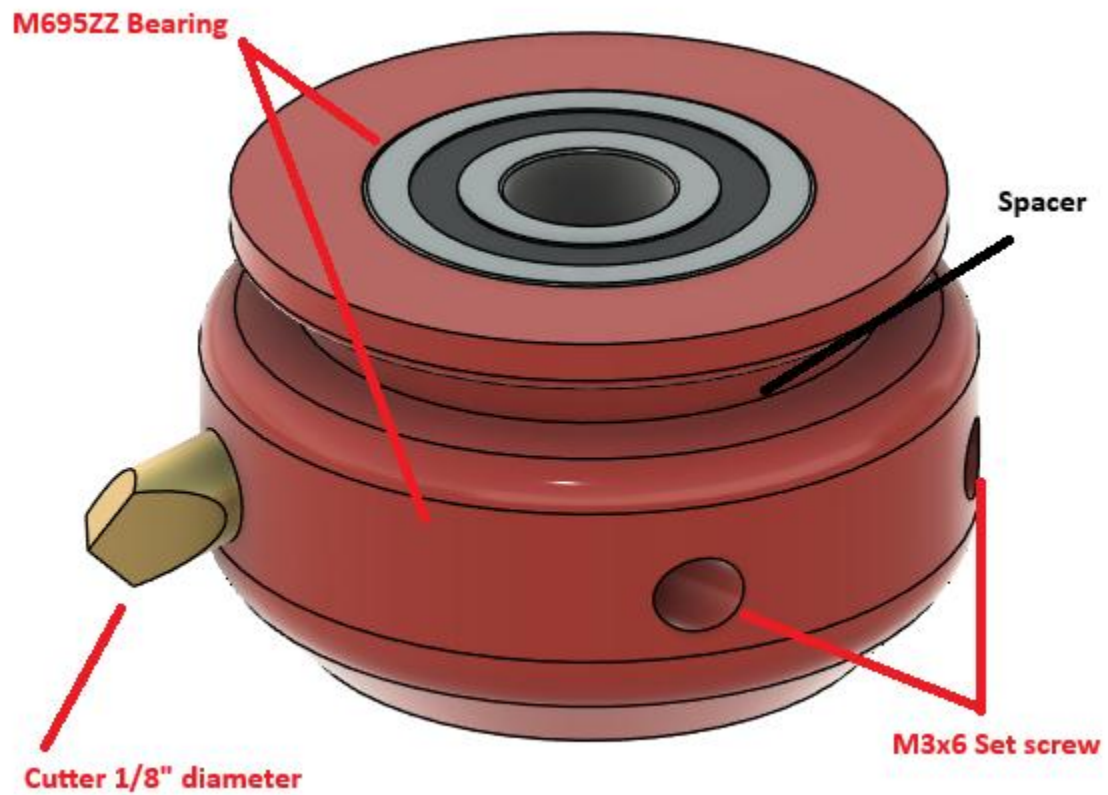
Cutter Heads

Insert Cutter Head



Use ABS slurry (ABS and Acetone) to secure Top Lock and nut.

Round Cutter Head



Tap the holes for the M3 set screws. Depending on how well your printer is calibrated, you may need to drill or ream the hole for the cutter.